Welcome everyone ☺

$a \% b = a$ if $(b > a)$

$3 \% 5 = 3$

$4 \% 8 = 4$

Today's Agenda :-

- → Modulas Basics
- → Modulas Properties
- → Interesting Problem (asked in Google).

$$\text{int } x \simeq [-2^{31}, 2^{31}-1] = [-2\times10^{9}, 2\times10^{9}]$$

$$\text{long } x \simeq [-2^{63}, 2^{63}-1] = [-8\times10^{18}, 8\times10^{18}]$$

$N \% a$ = Remainder when N is divided by a.

$10 \% 4 = 2$, $13 \% 5 = 3$

Dividend

Divisor → 5) 13 (2 → Quotient

10

3 → Remainder

Dividend = Divisor × Quotient + Remainder

Remainder = Dividend − Divisor × Quotient

↓

Greatest Multiple of Divisor $\leq$ Dividend.

Q1) $150 \% 11 = 150 -$ Greatest Multiple of 11 ≤ 150

$= 150 - 143$

$= \boxed{7}$

Q2) $100 \% 7 = 100 -$ Greatest Multiple of 7 ≤ 100

$= 100 - 98$

$= \boxed{2}$

$$\begin{array}{r} 14 \\ 7\overline{)100} \\ \underline{7} \\ 30 \\ \underline{28} \\ 2 \end{array}$$

Q3) $-40 \% 7 = -40 -$ Greatest Mul of 7 ≤ -40

$-40 - (-42)$

$= \boxed{2}$

Q4) $-60 \% 9 = -60 -$ Greatest Mul of 9 ≤ -60

$= -60 - (-63)$

$= \boxed{3}$

Python	C++ | Java | C | JS
$-40 \% 7 = 2$	$-40 \% 7 = -5 + 7 = 2$
$-60 \% 9 = 3$	$-60 \% 9 = -6 + 9 = 3$
$-10 \% 3 = 2$	$-10 \% 3 = -1 + 3 = 2$

Dividend → a, Divisor → M

```
a % M
if (a < 0) {
a % M =
  a % M + M
}
```

$40 \% -7 = 40 -$ Greatest Mul of (-7) $< = 40$ $(-7 * -5) = 35$

$\parallel$

$40 \% 7 = 40 - 35$

$= 5$

$\sim \% 10 = [0, 9]$

Why %? → limit our data to a required range

Eg Hashing Eg Consistent Hashing

Cryptography.

$$\left.\begin{matrix} -\infty \\ | \\ +\infty \end{matrix}\right\} \% M \Rightarrow [0, M-1]$$

Modular Arithmetic :-

$[0, M-1]$ $[0, M-1]$ $> (M) \% M =$ $[0, M-1]$

$[0, M-1]$

1) $(a+b) \% M = (a \% M + b \% M) \% M$

Eg $a = 6, b = 8, M = 10$

$(6+8) \% 10 = (6 \% 10 + 8 \% 10) \% 10$

$(14) \% 10 = (6 + 8) \% 10$

$4 = 14 \% 10$

$4 = 4$

3) $(a-b) \% m$

4) $\left(\frac{a}{b}\right) \% m$ (Adv)

2) $(a * b) \% M = (a \% M * b \% M) \% M$

$a = 6, b = 8, M = 10$

$(48) \% m = (48) \% m$

$8 = 8$

// No inbuilt functions.

// power(a, N, P) → Calculate $a^N \% P$.

$a = 2, N = 5, P = 7 \Rightarrow 2^5 \% 7 = 32 \% 7 = 4$

$a = 3, N = 4, P = 6 \Rightarrow 3^4 \% 6 = 81 \% 6 = 3$

```
power(a, N, P) {
    for(int i=1; i<=N; i++){
        a = a*a
    }
    return a % P
}
```

i		a
		$\underline{a}$
1	a = a*a	a^2
2	a = a*a	a^4
3	a = a*a	a^8
4	a = a*a	a^{16}

```
power(a, N, P) {
    ans = 1
    for(i=1; i<=N; i++){
        ans = a*a
    }
}
```

ans = 1 a = a

i		a
1	ans = a*a	a^2
2	ans = a*a	a^2
3	——	a^2
4	——	a^2
		⋮

Constraints :- $1 <= a <= 10^9$

$1 <= N <= 10^5$

$1 <= P <= 10^9$

TC - O(N)

```
int power(a, N, p) {  → a^N % p
    long ans = 1
    for(int i = 1; i<=N; i++) {
        ✓ans = (ans * a) % p }  ans ∈ [0, p-1]
        // ans = (ans % p * a % p) % p
    }
    return ans % p
}
```

$[0, p-1]$ $[0, p-1]$ $\simeq p^2 \simeq 10^{18}$ % p < P

Input :-

$a = 2, N = 40, ans = 2^{40}$

$a = 2, N = 100, ans = 2^{100}$

In for loop itself ans can overflow

(ans = ans * a)

$(a \% p) \% p = a \% p$

$(10 \% 3) \% 3 =$

$1 \% 3 = 1$

Dry Run a, N = 5, P → $a^5 \% p$ ans = 1

i	ans = (ans * a) % p	ans
1	ans = (1 * a) % p	a % p
2	ans = (a % p * a) % p	a^2 % p
3	ans = (a^2 % p * a) % p	a^3 % p
4	ans = (a^3 % p * a) % p	a^4 % p
5	ans = (a^4 % p * a) % p	a^5 % p

// Divisibility Rule of 3 ?

↳ (Sum of Digits) % 3 = 0 } Childhood

$1 \% 3 = 1$

$10 \% 3 = 1$

$10^2 \% 3 = 1$

$10^3 \% 3 = 1$

$10^n \% 3 = 1$

$(3458) = (3 \times 10^3 + 4 \times 10^2 + 5 \times 10^1 + 8 \times 10^0)$

$(3458) \% 3 = (3 \times 10^3 + 4 \times 10^2 + 5 \times 10^1 + 8 \times 10^0) \% 3$

$(3 \% 3 * (10^3 \% 3)) \% 3 = (3 \% 3) \% 3 = (3 \% 3)$

$((3 \times 10^3) \% 3 + (4 \times 10^2) \% 3 + (5 \times 10^1) \% 3 + (8 \times 10^0) \% 3) \% 3$

$\Rightarrow \boxed{(3 \% 3 + 4 \% 3 + 5 \% 3 + 8 \% 3) \% 3}$

$(a * b) \% m$

$(a * b \% m) \% m$

$((a \% m) * b) \% m$

// Divisibility Rule of 9 = (Sum of dig) % 9 = 0.

$1 \% 9 = 1$

$10 \% 9 = 1$

$10^2 \% 9 = 1$

$10^3 \% 9 = 1$

$\vdots$

$10^n \% 9 = 1$

// Divisibility Rule of 4

↳ (last 2 digits) % 4 == 0

$1 \% 4 \neq 0$

$10 \% 4 \neq 0$

$10^2 \% 4 = 0$

$10^3 \% 4 = 0$

$\vdots$

$10^x \% 4 = 0$

$x >= 2$

$(4328) = (4 \times 10^3 + 3 \times 10^2 + 2 \times 10^1 + 8 \times 10^0)$

$(4328) \% 4 = (4 \times 10^3 + 3 \times 10^2 + 28) \% 4$

$= ((\underbrace{4 \times 10^3) \% 4}_{0} + \underbrace{(3 \times 10^2) \% 4}_{0} + (28) \% 4) \% 4$

$= (28) \% 4$

Que :- Given a number in arr[] . Calculate number % p

Each arr[i] contains single digit of no.

Eg $N = 5$

0	1	2	3	4
3	2	4	6	4

$\Rightarrow (32464) \% 4 = 0$

$p = 4$

Constraints :- $1 <= N <= 10^5$

$1 <= p <= 10^9$

Idea 1) Convert arr[] into a number % p.

$N = 9 \Rightarrow 9$ digits $\simeq 10^9 \rightarrow$ int

$N = 18 \rightarrow 18$ digits $\simeq 10^{18} \rightarrow$ long

$N = 100 \rightarrow 100$ digits $= 10^{100} \rightarrow$ } ×

$N = 10^5 \Rightarrow 10^5$ digits $= 10^{10^5}$ } ×

Idea 2)

↳ (Divisibility Rules)

If $p = 2, 3, 4, 5, 6, 7, 8, 9, 10 \ldots 10^9$

X

	0	1	2	3	4	5	
arr[6] :-	3	2	8	4	2	9	o/p

$$(3\times10^5 + 2\times10^4 + 8\times10^3 + 4\times10^2 + 2\times10^1 + 9\times10^0)$$ o/p

i	pow
0	N-1 ⇒ N-(i+1)
1	N-2 N-i-1
2	N-3

// Generalization :-

arr[N] :-

0	1	2	. . . . .	N-1

$$\left[\left(ar[0] * 10^{N-1}\right) + ar[1] \times 10^{N-2} + ar[2] * 10^{N-3} + \ldots\ldots \left(ar[N-1] * 10^{0}\right)\right] \% p$$

$$\left[\left(ar[0] * 10^{N-1}\right) \% p + \left(ar[1] * 10^{N-2}\right) \% p + \left(ar[2] * 10^{N-3}\right) \% p + \ldots\ldots \left(ar[N-1] * 10^{0}\right) \% p\right] \% p$$

$$\left(ar[0] * 10^{N-1}\right) \% p$$

$$\Rightarrow \left(ar[0] \% p * 10^{N-1} \% p\right) \% p$$

Math.pow(10, N-1) % P → overflow

power(10, N-1, p) $\Rightarrow 10^{N-1} \% p$

i^{th} — $\left(ar[i] * 10^{N-i-1}\right) \% p$

$= \left[ar[i] \% p * power(10, N-i-1, p)\right] \% 1$

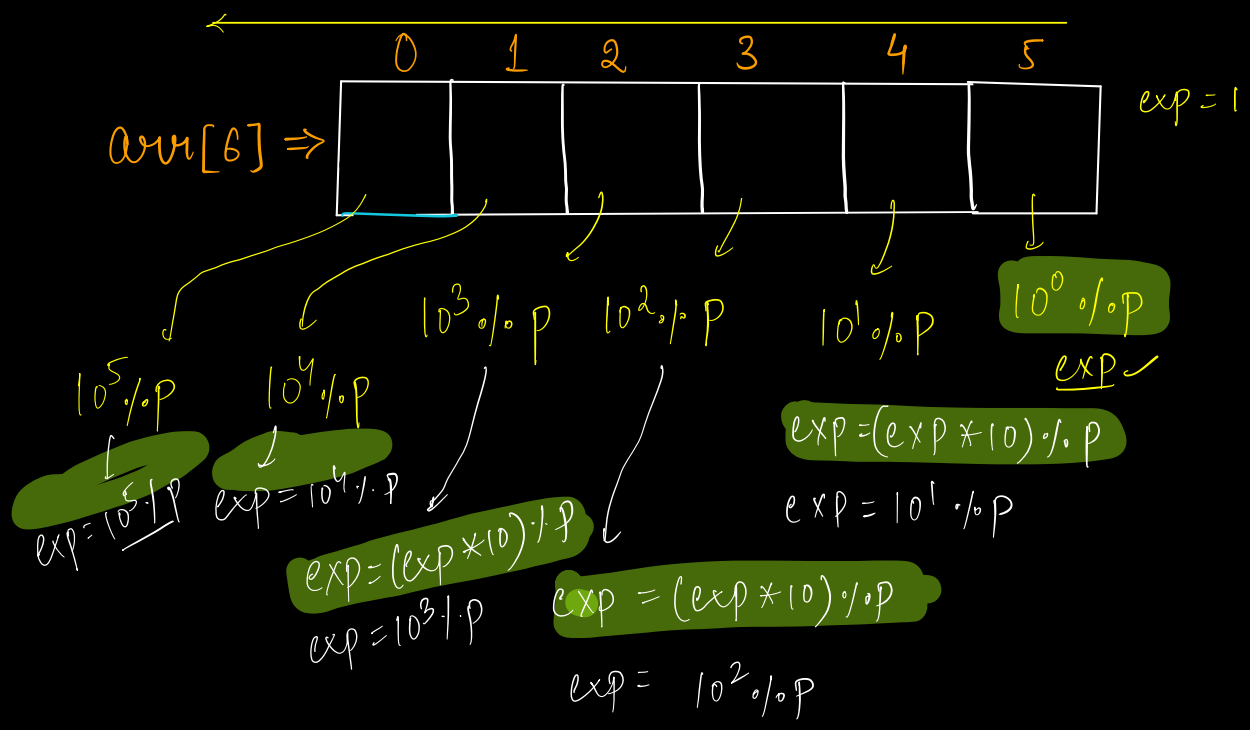
```
ans = 0
for(int i=0; i<N; i++){
    ans = ans + (ar[i]%p * power(10, N-i-1, p)] %p
    ans = ans %p
}
return ans
```

$O(N)$

$10^{N-i-1} \% p$

TC - $O(N^2)$

|| Optimization :-



— Calculate exp from R → L.

```
long ans = 0
long exp = 1
for(int i = N-1 ; i >= 0; i--){
    ans = [ans + (arr[i] % p * exp] % p] % p
    //ans = ans % p
    exp = (exp * 10) % p
}
return ans
```

TC - O(N)
SC - O(1)